

Jasim Kareem Badawi

SENIOR AI & SOFTWARE ARCHITECT | SECURITY RESEARCHER

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PROFESSIONAL SUMMARY

Highly technical, self-taught Software Architect and Security Researcher specializing in distributed systems, security-hardened fintech integrations, neobanking APIs, and financial guardrails for autonomous AI systems. Proven track record of building production-ready proxies, visual messaging automation engines, and secure transaction ledgers. Expert in security auditing, cryptographic implementations (AES-GCM, HMAC), and application security (AppSec) remediation. Combines deep technical execution (Node.js, TypeScript, Go, Python, FastAPI, Docker, Cloudflare Workers) with strong analytical communication to deliver robust, audit-compliant engineering solutions.

TECHNICAL SKILLS

AI & AGENTIC SYSTEMS:	AI Agents, LLM Fine-tuning & Edge Integration, Prompt Engineering, RAG Architectures, Custom Proxy Guardrails
FINTECH & CRYPTOGRAPHY:	Atomic Transactions, HMAC SHA-256 Webhook Verification, AES-256-GCM client encryption, PBKDF2, JWT/OAuth2, API Hashing
BACKEND & SYSTEMS:	Node.js, TypeScript, Go, Python, FastAPI, Cloudflare Workers, Hono, RESTful APIs, WebSockets, Redis, BullMQ, Docker
FRONTEND & DATABASES:	React, Vite, Tailwind CSS, shadcn/ui, PostgreSQL, Supabase, Prisma ORM, Database performance tuning & Row-Level Security (RLS)

FEATURED PROJECTS

Agmoney: Real-Time Financial Guardian Proxy for AI Agents

agmoney.aiandthings.tech | [GitHub](#)

Architected and deployed a "Financial Firewall" proxy sitting between autonomous AI agents and LLM APIs to enforce hard budget caps in real-time.

- Developed a low-latency edge gateway using Cloudflare Workers and Hono, maintaining proxy overhead under 15ms with full streaming support.
- Secured OpenAI Master Keys on the client side using AES-256-GCM symmetric encryption derived via PBKDF2.
- Engineered atomic cost calculation and database syncing using `ctx.waitUntil()` on Cloudflare Workers and Supabase (PostgreSQL).

SumerSend: Enterprise Messaging & Secured Payments Engine

sumersend.aiandthings.tech | [GitHub](#)

Developed an enterprise visual messaging builder featuring custom DNS validation (SPF/DKIM/MX), direct email delivery, and secure wallet billing.

- Integrated regional payment gateways (Zain Cash, AsiaHawala, FastPay, Qi Card), validating webhooks via HMAC-SHA256 signature matching.
- Audited application security, patching JWT session leakage, Stored XSS, and Row-Level Security (RLS) bypasses.
- Mitigated wallet double-spend race conditions by implementing atomic database transactions with row-level locking (`SELECT FOR UPDATE`).
- Orchestrated bulk off-thread campaign distribution via Redis-backed BullMQ queues.

Bank of Agents: API-First Neobanking Infrastructure for AI

bank.aiandthings.tech | [GitHub](#)

Built a neobanking prototype enabling autonomous AI agents to hold digital wallets, check balances, and spend within predefined rules.

- Implemented secure, low-latency API verification endpoints using hashed agent signature tokens.
- Programmed a Human-in-the-Loop (HITL) dashboard allowing users to review, approve, or reject pending agent transactions in real-time.
- Structured complex database schemas and transactional tables using Prisma ORM with PostgreSQL.

SECURITY RESEARCH & APPSEC AUDITING

Independent Security Researcher & Vulnerability Auditor

JUNE 2025 – PRESENT

- Facebook Privacy Boundary Bypass:** Discovered, documented, and reported a privacy boundary bypass vulnerability on the Facebook platform, creating a comprehensive Proof of Concept (PoC) and technical mitigation report.
- OTP Protocol Analysis:** Developed custom Python fuzzers and protocol checkers to analyze and identify logic vulnerabilities in SMS OTP verification endpoints.
- Security Tooling:** Built lightweight, automated application security scanners (`wp_advanced_scanner.py`, `wp_xss_deep_verify.py`) in Python to automate detection of SQL injections, cross-site scripting (XSS), and privilege escalation.

PROFESSIONAL EXPERIENCE

Independent Software Developer & Freelancer Remote	JULY 2023 – PRESENT
• Design, build, and deploy production-grade full-stack applications and secure APIs for remote international clients. • Conduct code reviews, database schema design, and query optimization for client databases (PostgreSQL/Supabase). • Write clear, highly technical developer guides, API specifications, and architectural documentation.	
Project Manager Private Family Library	SEPTEMBER 2024 – OCTOBER 2025
• Supervised business operations, managed book inventory systems, and coordinated team logistics. • Designed and maintained a digital custom inventory tracking system that reduced item mismatches by 18%.	

EDUCATION

Bachelor of Arts (BA) in English Language University of Basrah	JULY 2023
Self-Taught Computer Science & Security Engineering Independent Study (LLM infra, Fintech cryptography, AppSec)	CONTINUOUS